First Automations & Agent Concepts – Final Project

English translation of the original German document

Project Briefing

Business Case & Problem Statement

Volta Coffee Roasters in Porto ships around 20 wholesale orders per week to cafés and restaurants throughout Portugal and Spain. Operations Manager Inês faced a massive bottleneck: orders arrived via web forms, emails and WhatsApp voice notes, which meant she had to manually read and interpret every order and confirmation and type it into the system.

Goal & Solution Concept

An automated n8n hybrid workflow that handles routine orders, forwards ambiguous cases to Inês, and ensures that no customer communication takes place without her explicit approval (human-in-the-loop).

Workflow Architecture

- **Initial Extraction:** The webhook receives the payloads; the Google Gemini Chat Model with the Structured Output Parser extracts the key data fields.
- **Product Matching & Preparation:** The data is cleaned, set, looked up in Airtable and consolidated via JavaScript.
- **AI Agent Classification:** Categorizes orders as `standard`, `needs_review` or `needs_clarification`.
- **Rule-Based Branching:**
  - **Standard:** Logs the case in Airtable, notifies Slack and drafts a confirmation email via Gemini.
  - **Review/Clarification:** Logs in Airtable and triggers immediate Slack escalation alerts for Inês.
- **Approval Process:** The interactive Slack approval node pauses standard orders; approval (`true`) sends the Gmail confirmation, while a rejection (`false`) alerts the team.

Success Criteria

- Complete elimination of manual routine data entry.
- Prevention of automated errors or hallucinations through approval loops.
- Inês retains full decision-making authority at every checkpoint.

I-P-O Design with AI/Logic Tags

Inputs	Process	Outputs
Source / Channel: Webhook. **Field Shape:** JSON structure with attributes such as `id`, `category`, `expected_classification`, `channel`, `from_email`, `received_at`, `subject` and `body`. **Trigger Condition:** Incoming webhook request that starts the automated pipeline.	**Processing (8 steps, one verb each)** **1. Extract:** Extracting structured order data (such as customer ID, company name and order details) from the raw text using the Google Gemini Structured Output Parser. **2. Set:** Assigning and preparing the variables (`customer_id`, `raw_order_text`, `order_id`) for the downstream nodes. **3. Search:** Querying the product database in Airtable (Search records) to match items and stock levels. **4. Consolidate:** Merging the context and generating deterministic flags (e.g. stock level, validation rules) in the JavaScript node (Code in JavaScript). **5. Classify:** Analyzing the data with the AI Agent to determine the classification (`standard`, `needs_review` or `needs_clarification`). **6. Route:** Controlling the data flow through the Switch node based on the evaluated routing rules. **7. Log & Notify:** Creating or updating the record in Airtable and sending a notification to the Slack channel (notification 1). **8. Draft & Wait:** Generating an email draft via Gemini (draft email) and pausing the workflow until Inês gives interactive approval via Slack (Send message and wait for response).	**Destination:** Airtable (database), Slack (team channels & escalation/approval interfaces) and Gmail. **Format:** Structured records in Airtable, formatted Slack notification cards, HTML email drafts, escalation alerts for edge cases and ambiguous cases, and final order confirmations. **When it Fires:** Fires fully automatically when an order arrives via webhook; standard cases proceed to the email draft and human approval (`approved: true`), while edge cases (`needs_review`) and ambiguous cases (`needs_clarification`) immediately send escalation alerts to Slack for manual handling.

Validation Protocol

Test Run 1: SAMPLE-01 (Happy-Path Input)

- **Input:** `SAMPLE-01`
- **Intake:** Yes – The webhook successfully received the complete JSON payload.
- **LLM step:** Yes – Gemini produced valid JSON, with all required fields (`company_name`, `customer_id`, `order_details`, `id`) populated correctly.
- **Airtable:** Yes – The record was cleanly created in the "Wholesale Orders" table with matching fields.
- **Gmail draft:** Yes – A draft was created via the `draft email` node for pedro@cafeetlantico.pt with a sensible, personalized confirmation text.
- **Slack message:** Yes – The notification was posted in the `#wholesale-review` channel with the correct order context and approval buttons.
- **Conclusion:** The workflow handled this flawlessly (handled this OK).

Test Run 2: SAMPLE-04 (Edge-Case Input / Borderline Case)

- **Input:** `SAMPLE-04`
- **Intake:** Yes – The webhook received the payload without problems.
- **LLM step:** Yes – Gemini returned valid JSON, precisely extracted the domain-based company name (`cafelumriere.pt` → Café Lumière) and handled quantity/item variations.
- **Airtable:** Yes – The row appeared in Airtable with correctly mapped fields.
- **Gmail draft:** N/A – Based on the classification as an edge case (`needs_review`), it was routed to the review branch, which avoided an automated direct send.
- **Slack message:** Yes – An escalation alert was posted in Slack detailing the review request for Inês.
- **Conclusion:** The workflow handled this flawlessly.

Test Run 3: SAMPLE-08 (Ambiguous Input)

- **Input:** `SAMPLE-08`
- **Intake:** Yes – The payload was successfully ingested via webhook.
- **LLM step:** Yes – Valid JSON returned; unclear text was analyzed safely without triggering schema or runtime errors.
- **Airtable:** Yes – The record was created under the clarification status.
- **Gmail draft:** N/A – Email creation was bypassed, as expected for ambiguous inputs.
- **Slack message:** Yes – A clarification alert with context notes for team intervention was posted in Slack.
- **Conclusion:** The workflow handled this properly (had issues with input clarity, but did not crash).

Final Decision

- **Gate decision: PASSED (PASS)**
- **Summary:** All three test runs executed without pipeline crashes. The happy path was completed end-to-end through all target systems, while edge cases and ambiguous cases were safely routed to Slack escalation branches.

List of Limitations & Failure Modes

Unhandled Special Requests & Product Substitutions

- **Failure description:** If a customer requests a modification or custom substitution (e.g. "swap one of the bags for a fruity single-origin this week", as seen in SAMPLE-04), the current deterministic system has no rule designed to process or evaluate this change.
- **Affected workflow step / node:** LLM Parse & JavaScript step (Code in JavaScript).
- **Impact:** The system may misinterpret or partially omit the custom instruction, treating the order rigidly and risking an inaccurate confirmation.

Ignored Urgency Words

- **Failure description:** Time-critical signals in customer messages (e.g. phrases indicating imminent stock depletion) are completely ignored by strict deterministic rules.
- **Affected workflow step / node:** Classification & routing step (Classify / Switch node).
- **Impact:** Priority or urgent orders are treated identically to routine orders, without being flagged or prioritized for operational review.

Vague Reorders Without Details (Customers' "The Usual")

- **Failure description:** If a customer submits an ambiguous request saying "the usual" without specifying quantities or items (as demonstrated in SAMPLE-08), the system lacks the contextual order entry and the item details required for processing.
- **Affected workflow step / node:** Customer matching & parsing step (Customer Lookup / LLM Parse).
- **Impact:** Critical order details are missing, leading to incomplete parsing and requiring manual intervention.

Unusual Quantity Spikes or Event Orders

- **Failure description:** If an order volume deviates significantly from a customer's typical baseline (such as a massive peak for an event), rigid rules may still rate it as a standard routine order.
- **Affected workflow step / node:** Rule-based branching step (Switch node / Routing).
- **Impact:** Volume anomalies pass through the standard automation path unflagged, instead of being safely escalated for management review.

Agent Layer Justification

Context-Aware Reasoning

- **The problem:** Deterministic rules fail on linguistic nuances such as custom additions, implicit urgency or unclear wording.
- **The solution:** The AI agent evaluates the order content and deterministic indicators together to decide between automation and manual review.

Intelligent Routing

- **Standard (route standard):** Fully automated email draft creation and approval for routine orders.
- **Review required (route review):** Escalates special cases such as stock-level boundaries or product substitutions to Slack.
- **Clarification required (route clarification):** Triggers a Slack alert for incomplete or ambiguous orders.
- **Reasoning (agent_reason):** Provides transparent logging entries for every decision in Airtable and Slack.

Operational Value

- **Automation:** Automatic background processing reduces the team's daily workload.
- **Risk management:** Catches faulty or incomplete orders before emails are sent.

Test Plan & Results Protocol

- **Sample ID:** SAMPLE-01
- **Expected classification:** Standard
- **Actual classification:** Standard
- **Expected:** Gmail draft and Slack message triggered
- **Actual:** Gmail draft and Slack message triggered
- **Agent's reason (from the output):** All ordered items are in stock, are within the usual quantity limits and contain no urgency or substitutions.
- **Result:** Matches expectation

- **Sample ID:** SAMPLE-02
- **Expected classification:** Standard
- **Actual classification:** Standard
- **Expected:** Gmail draft and Slack message triggered
- **Actual:** Gmail draft and Slack message triggered
- **Agent’s reason (from the output):** All items are in stock, no urgency or substitutions were requested, and the ordered quantity is within the customer’s usual pattern.
- **Result:** Matches expectation

- **Sample ID:** SAMPLE-03
- **Expected classification:** Standard
- **Actual classification:** Standard
- **Expected:** Gmail draft and Slack message triggered
- **Actual:** Gmail draft and Slack message triggered
- **Agent’s reason (from the output):** The order meets all standard criteria, as all items are in stock, no urgency or substitutions were requested, and quantities are within the usual parameters.
- **Result:** Matches expectation

- **Sample ID:** SAMPLE-04
- **Expected classification:** Needs_Review
- **Actual classification:** Needs_Review
- **Expected:** No Gmail draft, but Slack message triggered
- **Actual:** No Gmail draft, but Slack message triggered
- **Agent’s reason (from the output):** The order requires review because the stock check showed partial item availability.
- **Result:** Matches expectation

- **Sample ID:** SAMPLE-05
- **Expected classification:** Needs_Review
- **Actual classification:** Needs_Review
- **Expected:** No Gmail draft, but Slack message triggered
- **Actual:** No Gmail draft, but Slack message triggered
- **Agent’s reason (from the output):** The order requires review because the requested items are only partially in stock.
- **Result:** Matches expectation

- **Sample ID:** SAMPLE-06
- **Expected classification:** Needs_Review
- **Actual classification:** Needs_Review
- **Expected:** No Gmail draft, but Slack message triggered
- **Actual:** No Gmail draft, but Slack message triggered
- **Agent’s reason (from the output):** This order requires review because items are only partially in stock and substitutions are present.
- **Result:** Matches expectation

- **Sample ID:** SAMPLE-07
- **Expected classification:** Needs_Clarification
- **Actual classification:** Needs_Clarification
- **Expected:** No Gmail draft and no Slack message triggered
- **Actual:** No Gmail draft and no Slack message triggered
- **Agent’s reason (from the output):** No output
- **Result:** Crashed

- **Sample ID:** SAMPLE-08
- **Expected classification:** Needs_Clarification
- **Actual classification:** Needs_Clarification
- **Expected:** No Gmail draft, but Slack message triggered
- **Actual:** No Gmail draft, but Slack message triggered
- **Agent's reason (from the output):** The order record is missing order details and item information, i.e. critical order details required for processing are missing.
- **Result:** Matches expectation

- **Sample ID:** SAMPLE-09
- **Expected classification:** Needs_Clarification
- **Actual classification:** Needs_Clarification
- **Expected:** No Gmail draft and no Slack message triggered
- **Actual:** No Gmail draft and no Slack message triggered
- **Agent's reason (from the output):** No output
- **Result:** Crashed

Written Summary & Project Reflection

- **Efficiency vs. control:** Integrating approval loops in Slack ensures that no unchecked draft leaves the company, while keeping the day-to-day manual input effort to a minimum.
- **Limits of deterministic logic:** Rigid rules quickly reach their limits with unstructured messages. Only the addition of the AI agent allows a context-aware evaluation of nuances such as local colloquial language (e.g. Portuguese "da casa"), quantity outliers or urgency.
- **Test insights & resilience:** The test protocol showed that the workflow performs as expected for standard and escalation cases. The crashes at SAMPLE-07 and SAMPLE-09 were caused by the unavailability of the external AI model service and not by errors in the workflow logic itself. For future iterations, setting up automatic retry logic or fallback nodes for API outages is therefore recommended to ensure seamless system stability.